



# Flextherm Eco G2 D

that's excellence.



**ENG** Installation and operating manual

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# 1. Introduction

## 1.1. General

The following instructions provide guidance for the installer and user of Flextherm Eco G2 D Heat Batteries.

The installation must be carried out by a competent installer in accordance with local codes and regulations for plumbing, electrical installations and potable water supply.

Please note that Flextherm Eco G2 D Heat Batteries are intended to be used with external heat sources via an Flextherm Eco G2 D heat pump key.

## 1.2. Symbols used

In these instructions the following symbols are being used to draw the user's attention to information of particular importance.



### **Warning**

Indicates a hazardous situation that, if not avoided, could result in death or serious injury.



### **Caution**

Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury or material damage.

### **Notice**

Signals information that is considered important but not hazard related.

## 1.3. Abbreviations

The following abbreviations are used in the manual,

- ASHP – Air Source Heat Pump
- BERV – Back-Expansion Relief Valve
- DHW – Domestic Hot Water
- DSR – Demand Side Response
- EV – Expansion Vessel
- GSHP – Ground-Source Heat Pump
- HP – Heat Pump
- HW – Hot Water
- PCBA – Printed Circuit Board Assembly
- PCM – Phase Change Material
- PRV – Pressure Reducing Valve
- TMV – Thermostatic Mixing Valve
- VIP – Vacuum Insulation Panel

Please note that the term External Heat Source in this manual can refer to compatible Ground Source (GSHP) & Air Source (ASHP) Heat Pump ranges, as well as boilers. Please visit <https://flamco.aalberts-hfc.com> or refer to document FTE-HP for an up-to-date list of compatible Ground Source & Air Source Heat Pump products. Please also refer to document FTE-PV for compatible Power Diverters when using Solar PV as an external heat source.

## 1.4. Responsibilities

### Manufacturer's Responsibilities

Our products are manufactured in compliance with the requirements of applicable EU & UK laws and regulations.

As an innovative company committed to achieving net zero, Flamco continuously improves its products, which means that all specifications and other information stated in this manual are subject to change without prior notice.

Our liability as the manufacturer may not be invoked in the following cases:

- Failure to abide by the instructions on using the Heat Battery.
- Faulty or insufficient maintenance of system components protecting the Heat Battery.
- Failure to abide by the instructions on installing the Heat Battery.

### Installer's Responsibilities

The installer is responsible for the installation and the commissioning of the Heat Battery. The installer must:

Ensure they are suitably qualified for the level of plumbing and electrical work required for the installation of this Heat Battery.

- Read, understand and follow the instructions given in the manuals provided with the Heat Battery.
- Comply with the prevailing legislation and standards when carrying out the installation.
- Perform the initial start-up and carry out any checks necessary.
- Complete the commissioning procedure and checklist in this manual.
- Explain the installation to the user.
- If maintenance on system components is necessary, warn the user of the requirement to check the system components to maintain the system in good working order.
- Give all the instruction manuals to the user.

### **User's Responsibilities**

To achieve optimum operation and longevity of the Heat Battery, the user must adhere to the following instructions:

- Read and follow the instructions provided in the manuals provided with the Heat Battery.
- Call on qualified professionals to carry out installation, initial start-up and commissioning.
- Ask your installer to explain your installation to you.
- Ensure that system components are maintained as necessary.
- Keep the instruction manuals in good condition and close to the Heat Battery.



#### **Caution**

Cleaning and user maintenance shall not be carried out by children. Children must be supervised to ensure they do not play with the Heat Battery.

## 2. Safety

### 2.1. General safety notes

**Warning**

Only competent persons suitably qualified to carry out plumbing and electrical work may undertake installations, repairs or relocations of the Heat Battery.

Product training on the full range of Flextherm Eco G2 D Heat Batteries is available from Flamco or authorised training partners. For more information visit <https://flamco.aalberts-hfc.com>.

**Caution**

Do not operate the immersion heater until all heat exchanger circuits have been filled and the plumbing has been appropriately commissioned.

**Caution**

A thermal regulating or cut-off device **MUST** be present on the External Heat Source to ensure that the flow temperatures to the Heat Battery **DO NOT** exceed 80°C.

**Warning (user)**

If this Heat Battery develops a fault, switch the Heat Battery off at the nearest isolator and contact the installer. Shut off the water supply to the Heat Battery if necessary.

There are no user serviceable, adjustable, or settable parts in this Heat Battery. Do not remove or adjust any component, cover, or part of this Heat Battery – contact your qualified installer.

**DO NOT** bypass the thermal cut-out(s) in any circumstances.

## 3. Product specifications

### 3.1. Technical specifications

|                                                          | Unit            | Flextherm Eco 6D G2 | Flextherm Eco 9D G2 | Flextherm Eco 12D G2 |
|----------------------------------------------------------|-----------------|---------------------|---------------------|----------------------|
| Water content Primary Circuit                            | L               | 3.7                 | 5.3                 | 6.4                  |
| Fresh water content Secondary Circuit <sup>1</sup>       | L               | 3.7                 | 5.3                 | 6.4                  |
| Equivalent Hot Water Cylinder Size <sup>2</sup>          | L               | 142                 | 212                 | 284                  |
| Volume of hot water available at 40°C (V40) <sup>3</sup> | L               | 199                 | 301                 | 402                  |
| Equivalent Hot Water Cylinder Size <sup>4</sup>          | L               | 128                 | 192                 | 256                  |
| Volume of hot water available at 40°C (V40) <sup>5</sup> | L               | 167                 | 271                 | 333                  |
| Heat loss                                                | kWh/<br>24h (W) | 0.67 /<br>(28.1)    | 0.77 / (32.1)       | 0.84 /<br>(34.9)     |
| Energy label class <sup>6</sup>                          | -               | A+                  | A+                  | A+                   |
| Recommended maximum charging flow rate                   | L/Min           | 15                  | 20                  | 25                   |
| Recommended maximum HW flow rate <sup>7</sup>            | L/Min           | 15                  | 20                  | 25                   |
| Minimum supply pressure at Heat Battery inlet            | MPa<br>(Bar)    | 0.15 (1.5)          | 0.15 (1.5)          | 0.15 (1.5)           |
| Recommended operating pressure/<br>PRV set point         | MPa<br>(Bar)    | 0.3 (3)             | 0.3 (3)             | 0.3 (3)              |
| Maximum operating pressure/PRV set point                 | MPa<br>(Bar)    | 0.5 (5)             | 0.5 (5)             | 0.5 (5)              |
| BERV recommended set point                               | MPa<br>(Bar)    | 0.6 (6)             | 0.6 (6)             | 0.6 (6)              |
| Maximum design pressure/BERV maximum set point           | MPa<br>(Bar)    | 1.0 (10)            | 1.0 (10)            | 1.0 (10)             |
| Maximum Heat source flow temperature <sup>8</sup>        | °C              | 80                  |                     |                      |
| Minimum Heat source return temperature <sup>9</sup>      | °C              | 63                  |                     |                      |
| Pressure loss characteristics                            | -               | See Figures 3 & 4   |                     |                      |
| Recommended TMV setting                                  | °C              | 45-55               |                     |                      |
| Connected load at ~ 230 V, 50Hz                          | W               | 2800                |                     |                      |
| Power supply   Standby consumption                       | W               | 1 PH AC 230 V   7   |                     |                      |

Table 1 - Flextherm Eco G2 D technical specifications



**Table 1 Notes:**

1. Water content of the Heat Battery for sizing expansion vessels.
2. Calculated from the storage capacity of the Heat Battery when charged to maximum set points and assuming that the equivalent hot water cylinder thermostat is set at 60°C, mains cold water inlet temperature is at 10°C and the stored energy utilisation factor of the cylinder is 0.85.
3. The hot water volume available from the Heat Battery normalised to an average outlet temperature of 40°C when it is fully charged by the back-up electric heating element.
4. Calculated from the storage capacity of the Heat Battery when charged to heat pump set points and assuming that the equivalent hot water cylinder thermostat is set at 60°C, mains cold water inlet temperature is at 10°C and the stored energy utilisation factor of the cylinder is 0.85.
5. The hot water volume available from the Heat Battery normalised to an average outlet temperature of 40°C when charged to heat pump set points.
6. When heated by an External Heat Source.
7. While the Heat Battery can deliver higher flow rates than those listed, doing so will result in reduced performance in terms of duration of discharge and energy provided.
8. DO NOT exceed this temperature value when charging the Heat Battery using an External Heat Source. A thermal regulating or cut-off device MUST be present on the external heat source to prevent this.
9. The External Heat Source MUST be able to reach this temperature on the Return back to the External Heat Source from the Heat Battery Outlet at the end of the charging cycle.

### 3.2. General overview

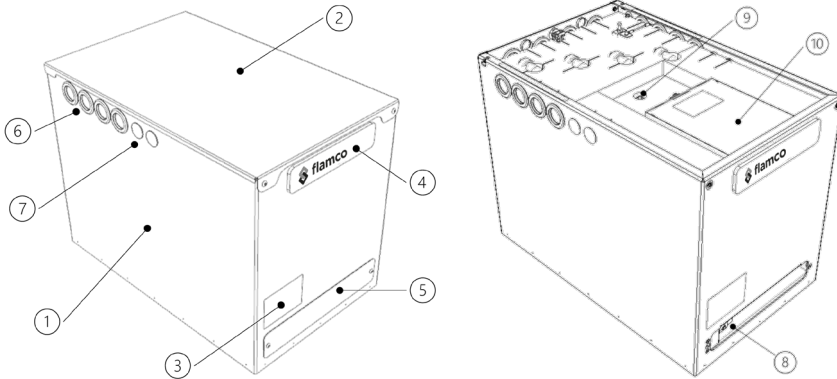


Figure 1 - General overview of the Flextherm Eco G2 D Heat Battery

| Item | Description                                                                 |
|------|-----------------------------------------------------------------------------|
| 1    | Heat Battery - Main body                                                    |
| 2    | Heat Battery - Lid                                                          |
| 3    | Heat Battery data badge / Serial number                                     |
| 4    | Controller interface                                                        |
| 5    | Heater terminal cover plate                                                 |
| 6    | Pipe entries (3 sides)                                                      |
| 7    | Cable entries (3 sides)                                                     |
| 8    | Non-self-resetting thermal cut-out (Remove Heater Terminal cover plate (5)) |
| 9    | Heat Battery Temperature Sensor Dry Pocket*                                 |
| 10   | Heat Battery Controller                                                     |

Table 2 - General overview of the Flextherm Eco G2 D Heat Battery

\*Do not insert temperature sensors other than those supplied with the Heat Battery.

### 3.3. Dimensions

#### General Dimensions

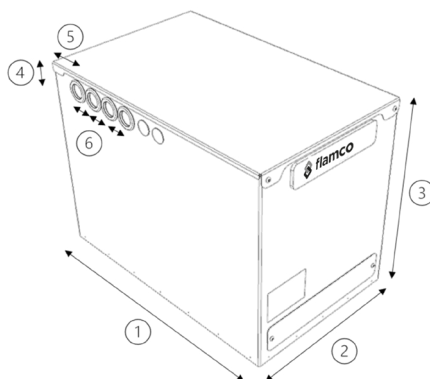


Figure 2 - Flextherm Eco G2 D Heat Battery dimensions

| All data in mm                             |                         | Flextherm Eco 6D G2 | Flextherm Eco 9D G2 | Flextherm Eco 12D G2 |
|--------------------------------------------|-------------------------|---------------------|---------------------|----------------------|
| 1 - Length                                 |                         | 575                 |                     |                      |
| 2 - Width                                  |                         | 365                 |                     |                      |
| 3 - Height                                 |                         | 640                 | 870                 | 1050                 |
| Centre of side pipe-entry from             | 4 -Top                  | 37                  |                     |                      |
|                                            | 5 - Rear                | 78                  |                     |                      |
|                                            | 6 - Centre of next pipe | 50                  |                     |                      |
| Centre of rear pipe-entry from (not shown) | Top                     | 37                  |                     |                      |
|                                            | Sides                   | 78                  |                     |                      |
|                                            | Centre of next pipe     | 70                  |                     |                      |

Table 3 - Flextherm Eco G2 D Heat Battery dimensions

### 3.4. Weights

| All data in kg      | Flextherm Eco 6D G2 | Flextherm Eco 9D G2 | Flextherm Eco 12D G2 |
|---------------------|---------------------|---------------------|----------------------|
| Gross weight        | 134                 | 181                 | 225                  |
| Net weight (empty)  | 129                 | 176                 | 220                  |
| Net weight (filled) | 136                 | 187                 | 233                  |

Table 4 - Flextherm Eco G2 D Heat Battery weights

### 3.5. Pressure loss

The Pressure Loss values detailed in Figure 3 are the pressure differences between the Cold-Water Inlet (Port D) and Hot Water Outlet (Port A) of the Heat Battery, Secondary (Fresh Water) Circuit.

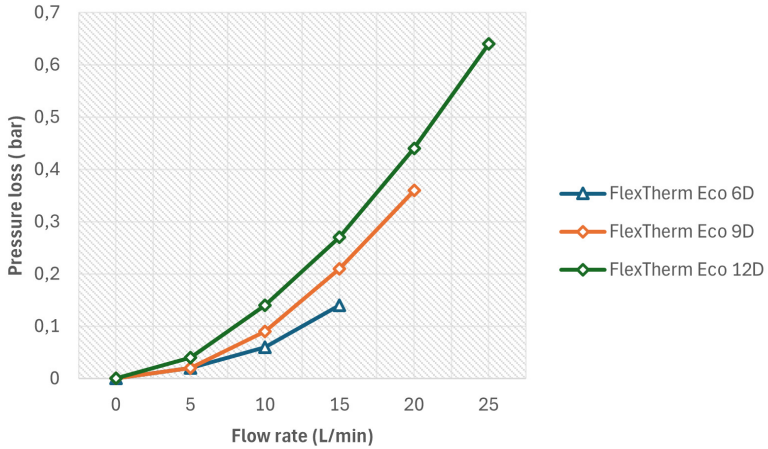


Figure 3 - Flextherm Eco G2 D Heat Batteries pressure loss - Secondary Circuit

The Pressure Loss values detailed in Figure 4 are the pressure differences between the External Heat Source Flow into Heat Battery (Port B) and External Heat Source Return from Heat Battery (Port C), Primary (Heat Source) circuit.

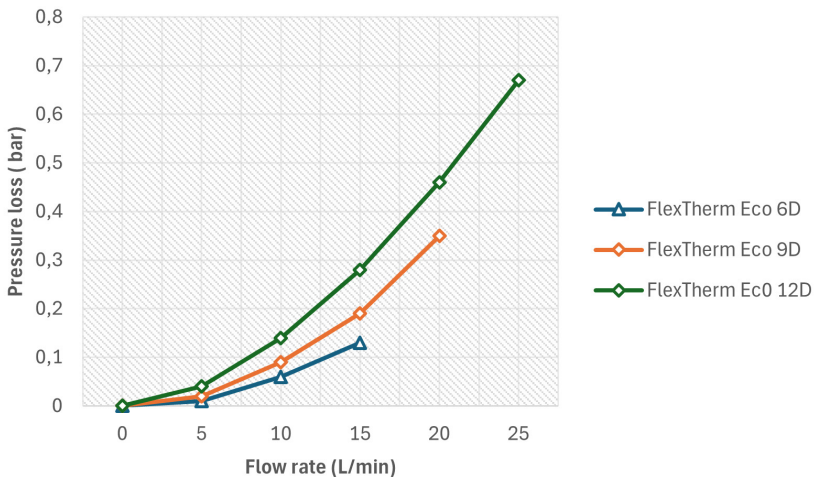


Figure 4 - Flextherm Eco G2 D Heat Batteries pressure loss - Primary Circuit

## 4. Product overview

### 4.1. General description

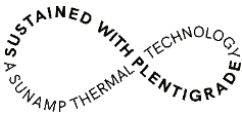
Flamco Flextherm Eco G2 D Heat Batteries are modern-day, energy-saving thermal stores made with a high-performance Phase Change Material (PCM) to deliver fast-flowing hot water reliably, safely, and efficiently. Up to four times smaller than the equivalent hot water cylinder, the sleek, super-compact design means the Flextherm Eco G2 D looks great in any home and frees up valuable storage space. They are also easy to install and require no mandatory annual maintenance.

Flextherm Eco G2 D Heat Batteries are charged by Air Source or Ground Source Heat Pumps or Gas Boilers and can be configured to work with electricity from the power grid as a back-up (Flextherm Eco G2 D heat pump -Key required - range HP-Key) or with excess power from a solar PV installation (Flextherm Eco G2 D heat pump -Key required - range PV+HP-key).

### 4.2. How it works

The secret to the success of Flamco Heat Batteries is the world leading, patented Plentigrade® technology. The Flextherm Eco G2 D range uses the high-performance, non-toxic and non-flammable Plentigrade P58 phase change material to deliver hot water on demand.

PCMs absorb, store and release large amounts of latent heat when changing state between solid and liquid. This unique formula stores up to four times more energy than water over the same temperature range, which means Flextherm Eco G2 D Heat Batteries are up to four times smaller than the hot water cylinders they replace.



The 'Sustained with 'Plentigrade' quality mark on our products is assurance of performance, efficiency, material safety, and reliability.

### 4.3. Intended use

Flamco Flextherm Eco G2 D Heat Batteries are intended for use as hot water appliances for domestic and residential uses only.

The product is intended for installation in a frost-free and weather-protected environment where it cannot be damaged by weather conditions.

Flextherm Eco G2 D products are designed to produce domestic hot water by using compatible ranges of Ground Source or Air Source Heat Pumps or Boilers as a main source of heat, these range of heat batteries have a standby heating element built in to act as a backup if the main heat source fails. The Heat Batteries can be configured for different use-cases by using an Flextherm Eco G2 D heat pump key.

Using a Flextherm Eco G2 D heat pump key from the HP-Key range will operate the product as an indirect water heater with optional electric back-up mode (back-up switch required). Applying a Flextherm Eco G2 D heat pump key from the PV+HP-key range, in combination with a power diversion controller, allows the use of surplus PV energy to be maximised, with the external heat source taking over when there is no PV electricity available at the time.

### 4.4. Storage & handling



#### **Warning**

Take into account the weight of the Heat Battery (Table 4) and local Health & Safety Regulations and Practices when considering safe lifting methods for moving the Heat Battery.

Verify all floors during transportation, storing or installation of the Heat Battery are suitable for the product weight (Table 4). The Heat Battery **MUST** not be stepped or sat on at any time during storage, handling, installation and use.



#### **Caution**

The Heat Battery must be stored in a dry, weather-protected and frost-free environment. The Heat Battery will incur damage if exposed to weather conditions including but not limited to rain, snow and extremes of temperature.



#### **Caution**

Do not tilt the product more than 45 degrees during the transportation or installation process.

## 5. Pre-installation



### Caution

Ensure that the following requirements have been considered and are met before choosing or installing a Flamco Flextherm Eco G2 D Heat Battery.

### 5.1. Water supply

- The Heat Batteries are not suitable tank fed hot water systems. Tank fed hot water systems have to be converted to mains pressurised systems when installing a Flamco Heat Battery.
- Ensure that the water supply requirements are within the Minimum and Maximum Pressure & Maximum Flow Rates detailed in (Table 1).
- Where mains water hardness can exceed 150 ppm Total Hardness, you must install a scale reducing device in the cold-water supply to the Heat Batteries.
- Limescale can be controlled using chemical limescale inhibitors, polyphosphate dosing, electrolytic scale reducers or water softeners (please refer to the manufacturer's instructions for servicing any water conditioning system).
- All system components used in the Heat Battery installation **MUST** be suitable for potable water and approved by local water regulations.
- The external heat source circuit (closed circuit) must be fitted with an appropriately sized expansion vessel and expansion relief valve.
- Use of the Heat Battery in combination with any additives to the secondary (fresh water) circuit from port D to A (with the exception of suitable water softeners in areas where water hardness exceeds 150ppm – please refer to above points), including dye, coolant or soldering flux, will invalidate the Heat Battery warranty and are not considered standard intended use.

### 5.2. Locating the heat battery



### Caution

The Heat Battery **MUST** be installed indoors and in a frost-free environment. Installations in locations such as unheated attics, garages etc may result in damage to the appliance and will affect your warranty. Please refer to the warranty terms and conditions and, if in doubt, contact Flamco customer support.

## Notice

Installation of the product at a height may affect the terms of your warranty. Please refer to the warranty terms and conditions and, if in doubt, contact Flamco customer support.

- The location of the Heat Battery installation **MUST** be in an indoor frost-free environment.
- Assess the location where the Heat Battery will be installed, for the spatial & clearance requirements of the Heat Battery **(Figure 5, Table 5)**.
- Ensure that the location chosen has a hard, solid and level surface that can withstand the weight of the Heat Battery as detailed in **(Table 4)**
- Ensure that the Heat Battery can be transported to the desired installation location, considering the weight of the Heat Battery and safe lifting methods according to local Health & Safety Regulations and Practices.
- If using multiple Heat Batteries, do not stack the Heat Batteries with internal controller directly on top of each other. Racking must be used to ensure access to the water connections and controller.

The following clearances should be considered:

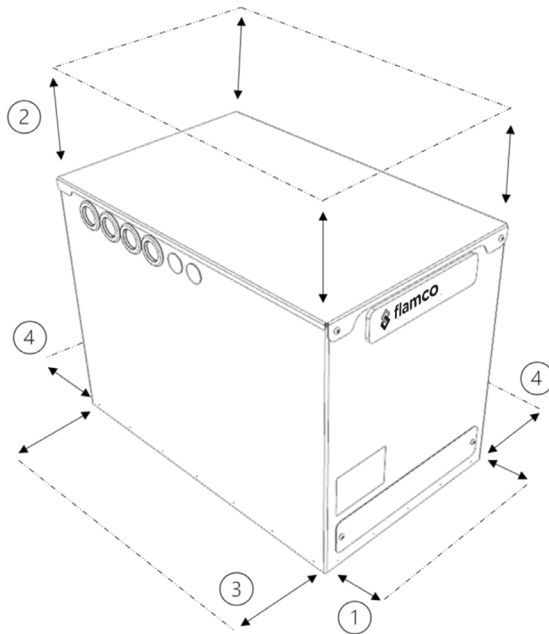


Figure 5 - Flextherm Eco G2 D Heat Battery Spatial Requirements



| Item | Distance | Notes                                                                               |
|------|----------|-------------------------------------------------------------------------------------|
| 1    | 150mm    | To access terminal cover and to ensure visibility of the data badge and LEDs        |
| 2    | 450mm    | To remove lid and access internals                                                  |
| 3    | 150mm    | To allow for pipe and cables entry and minimum cable-bend radiuses (side dependent) |
| 4    | 10mm     | If no access required (side dependent)                                              |
| -    | < 3000mm | Recommended length of cable runs                                                    |

Table 5 - Flextherm Eco G2 D Heat Battery Spatial Requirement

## 6. Installation

### 6.1. General



#### Warning

Before commencing the Hydraulic Installation of the Heat Battery you must ensure that the Heat Battery is electrically isolated from the Mains.



#### Caution

To avoid damage to the VIP insulation of the appliance do not:

- Carry out work that may leave sharp or abrasive residue in the Heat Battery, such as deburring of pipes, drilling of holes or stripping of wires above the open appliance.
- Deposit tools in the open appliance.
- Use sharp objects, such as cutters or similar, to cut grommets or insulation layers while they are located in the appliance.

#### Notice

Prior to installing the Heat Battery, please familiarise yourself with the product by looking at Figure 1 & Table 2 (general product overview) and ensure that all pre-installation requirements (Chapter 5) are met.

- Remove lid. The lid is secured by 2 x M5 button head cap screws at the front and two locating pins at the back (Figure 6):
  - o (1) Remove 2 x M5 button head cap screws using 3mm hex head and set aside.
  - o (2) Slide the lid forward, (3) then lift the lid up and set aside.

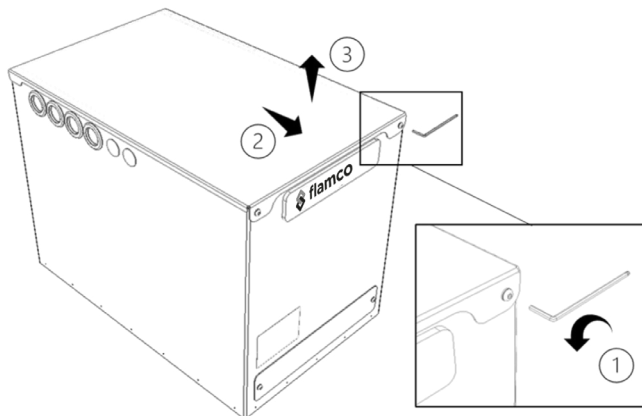


Figure 6 - Lid Removal of Flextherm Eco G2 D Heat Battery

## 6.2. Water connections



### Warning

All connection pipe work inside the Heat Battery casing **MUST** be using  $\varnothing$  22mm copper pipe. This is to allow the earth connection between the case, inlet, and outlet pipe fittings.



### Caution

All Plumbing components used for the Heat Battery installation **MUST** be approved for use with potable water according to local water regulations.

Do not fit any check valves between Heat Battery, BERV & Expansion Vessel.

The Heat Battery controller lid **MUST** be kept closed whilst performing any hydraulic works. This is to avoid water or particles coming into contact with the PCBA and other electrical components & wiring of the Heat Battery controller.

Do not carry out hot works on the appliance.

Please follow the instructions below in conjunction with Figure 8 for the Heat Battery hydraulic installation:

### Notice

Sizing of the pipe work must consider mains water supply pressure, the design flow rates, size of the Heat Battery and pressure loss detailed in Figure 3 & Figure 4.

Remove the top two layers of insulation (layer 1 is 10mm and layer 2 is 32mm thick) and set aside.

- Rotate the elbows to the side you wish to connect the hydraulics (left, right or back) (Figure 7).
- The cold water inlet must be connected to Port D & the hot water outlet to Port A. Meanwhile the External Heat Source Flow to Heat Battery must be connected to Port B & External Heat Source Return from the Heat Battery to Port C (Figure 7, left hand side).

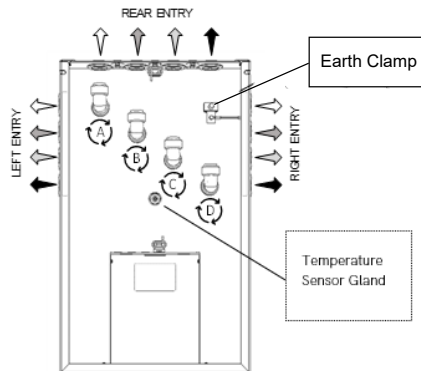


Figure 7 - Flextherm Eco G2 D Ports

- Remove the respective rubber grommets (Item 3 – Figure 10) in the outer housing and cut the centres (with a cross) with a knife. Re-insert the cut grommets. Do not cut the grommets in place, as this may damage the VIP.

#### **Notice**

It is recommended that the pipes exiting the Heat Battery are installed in a manner that avoids thermo-syphons, as this may increase the heat losses of the installation.

- Cut and prepare the Ø22mm copper pipe to suit the rest of the installation / system:
  - Always cut the pipe uniformly and at a 90 degree angle, using a rotary pipe cutter whenever possible. Ensure the cutter wheel is appropriate for the copper pipe.
  - Deburr the pipe end, both internally and externally to create a 1mm chamfer on the outside of the pipe.
  - Check the pipe ends are free from damage and clean, wiping away any swarf to avoid damaging the 'O' ring inside the push-fit elbow upon pipe insertion.
  - The pipe end must also be free from stickers, tape and adhesive residues.
  - Mark the socket depth (27mm) on the pipe with a marker.
  - Insert the pipe firmly with a slight twisting action until it reaches the pipe stop with a positive "click".
  - Ensure the depth insertion mark corresponds with the mouth of the fitting, then pull firmly on the pipe to ensure the fitting is secure.
- Fit the earth clamp to the Ø22mm copper pipe.
- Connect to the rest of the fixed system hydraulics.
- If carrying out hot works (such as soldering, welding or brazing) these must be performed on pipes detached from the Heat Battery (minimum 1 metre away).
- Fill the system with water, purging any air out of the system. This may take several minutes and can be aided by repeatedly opening and closing the outlet. This must be applied to both circuits of the Heat Battery.
- Once finished purging and with the system pressurised, inspect the piping/tubing and joints for any leaks. Take remedial action if necessary.

#### **Notice**

After completion of the leak checks or commissioning process, all connected pipes MUST be adequately insulated for at least 1m from their connection points with the Heat Battery to avoid increased heat losses through the connected pipework.

| Item | Description                                             | Notes                                                                                                                                                                       |
|------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Mains Cold Water Supply                                 |                                                                                                                                                                             |
| 2    | Mains Check Valve                                       | The Installer MUST inspect the presence of this in the property and install a BERV if present as detailed in (4).                                                           |
| 3    | Mains Cold Water Pressure Reducing Valve                | This MUST be fitted (please see Table 7 for more information).                                                                                                              |
| 4    | Mains Back-Expansion Relief Valve                       | This MUST be fitted. The maximum rating of the Valve MUST be no higher than 10 Bar (please see Table 7 for more information).                                               |
| 5    | Expansion Vessel                                        | This MUST be fitted. The Charge Pressure of the EV MUST be equal to the pressure setting of the PRV (item 3) (please see Table 7 for more information).                     |
| 6    | Cold Water Inlet - Port D                               | MUST be 22mm OD Copper pipe                                                                                                                                                 |
| 7    | Hot Water Outlet - Port A                               | MUST be 22mm OD Copper pipe                                                                                                                                                 |
| 8    | Hot Water Tempering Valve                               | This MUST be fitted & regulated to provide hot water output ranging between 45°C and 55°C.                                                                                  |
| 9    | Hot Water Flow Regulator Valve                          | Set the flow rate at the outlet of the Heat Battery to match the maximum recommended flow rate for the relevant Heat Battery size (Table 1).                                |
| 10   | Hot Water Isolation Valve                               |                                                                                                                                                                             |
| 11   | Hot Water Supply to Dwelling                            |                                                                                                                                                                             |
| 12   | Cold Water Supply to Dwelling                           |                                                                                                                                                                             |
| 13   | Heat Source (eg Air Source Heat Pump)                   | Boilers, Air Source & Ground Source Heat Pumps (Please check document D0086 for list of compatible heat pump ranges).                                                       |
| 14   | External Heat Source Flow - Port B                      | MUST be 22mm OD Copper pipe                                                                                                                                                 |
| 15   | External Heat Source Return - Port C                    | MUST be 22mm OD Copper pipe                                                                                                                                                 |
| 16   | Auto-bypass Valve                                       | This MUST be fitted & regulated to meet Flow Rate requirements and Heat Pump's defrost minimum flow rate and volume requirements (please see Table 7 for more information). |
| 17   | External Heat Source Central Heating/DHW Diverter Valve | Ensure that a Diverter Valve is used and not a Mid-Position Valve                                                                                                           |
| 18   | Central Heating Flow                                    | Refer to External Heat Source Manufacturer Instruction's for sizing and installation for this part of the system                                                            |
| 19   | Central Heating Return                                  | Refer to External Heat Source Manufacturer Instruction's for sizing and installation for this part of the system                                                            |

|    |                              |                                                                              |
|----|------------------------------|------------------------------------------------------------------------------|
| 20 | Anti-thermosiphon pipework   | Recommended if the pipework from the appliance goes horizontally or upwards. |
| 21 | Heat Battery Isolation Valve | This MUST be fitted (please see Table 7 for more information).               |
| -  | Pipework Insulation          | All exposed pipework should be adequately insulated                          |

Table 6 - Figure 8 diagram descriptions

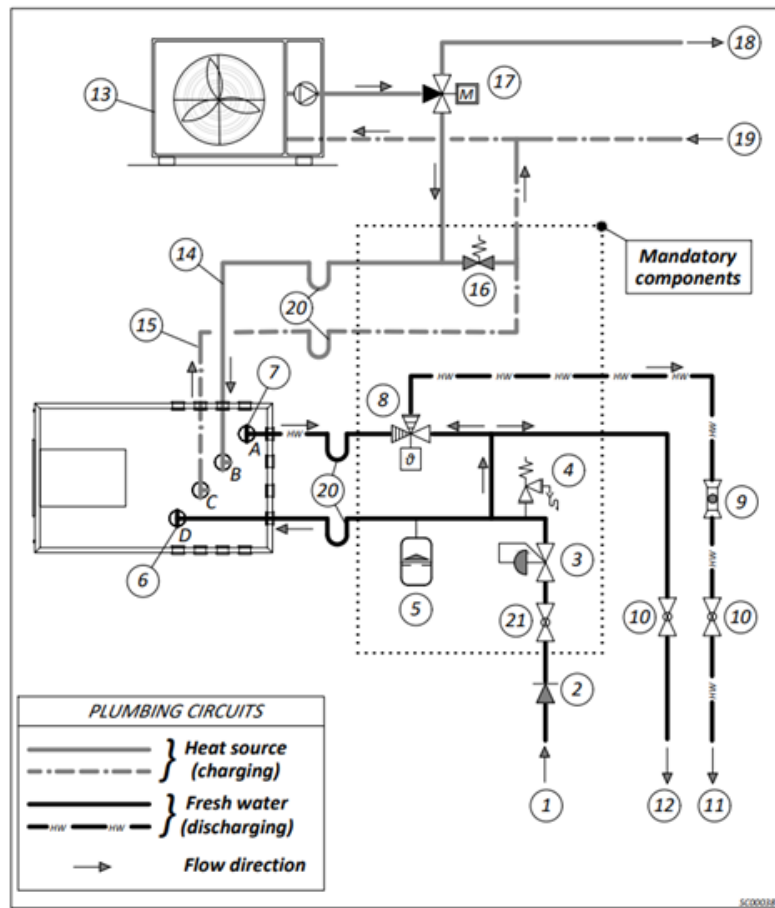


Figure 8 - Flextherm Eco G2 D plumbing diagram

## 6.3.Mandatory Plumbing Components



### Caution

The components within the dotted border in Figure 8 **MUST** be fitted with each Heat Battery installation (NOT fitting them may lead to Heat Battery damage and invalidation of the warranty) .

The BERV can be located away from the Heat Battery, provided there are no check valves between the BERV and the Heat Battery. The BERV can be discharged in existing internal or external drainage according to local codes and regulations.

The following plumbing components are mandatory and are required for the Heat Battery warranty to be valid (items 3, 4, 5, 8, 16 & 21 **MUST ALWAYS** be fitted. The remaining items must be fitted in certain circumstances – please refer to the Notes. Please also refer to the manufacturer's instructions for the maintenance of these components):

| Item | Description                              | Notes                                                                                                                                                                                                                                                                                                                                                                                   |
|------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | Mains Cold Water Pressure Reducing Valve | The maximum rating of the Valve <b>MUST</b> not exceed the maximum operating pressure of the Heat Battery (see Table 1).                                                                                                                                                                                                                                                                |
| 4    | Mains Check Valve                        | The maximum rating of the Valve <b>MUST</b> be no higher than the maximum design pressure of the Heat Battery (10 Bar).                                                                                                                                                                                                                                                                 |
| 5    | Expansion Vessel                         | Charge Pressure of Expansion Vessel <b>MUST</b> be equal to the pressure setting of the PRV (item 3). A minimum of 0.5L expansion vessel is required (please follow the correct calculation methodology for sizing EVs). The charge pressure must be checked & topped up in line with the expansion vessel manufacturer's instructions for maintenance or annually, whichever is sooner |
| 8    | Hot Water Tempering Valve                | <b>MUST</b> be fitted & regulated to provide hot water output ranging between 45°C and 55°C .                                                                                                                                                                                                                                                                                           |
| 16   | Auto-bypass Valve                        | <b>MUST</b> be regulated to meet Flow Rate requirements and Heat Pump's defrost minimum flow rate and volume requirements (if external heat source has an integrated auto-bypass valve, please do not install another!).                                                                                                                                                                |
| 21   | Heat Battery Isolation Valve             | <b>MUST</b> be fitted in order to enable safe & adequate Heat Battery maintenance (if required).                                                                                                                                                                                                                                                                                        |
|      | Water Conditioner                        | <b>MUST</b> be installed in areas where water hardness can exceed 150ppm                                                                                                                                                                                                                                                                                                                |

Table 7 - Mandatory plumbing components

## 6.4. Electrical connections



### **Warning**

All Electrical wiring should be carried out by a competent person and be in accordance with the latest local wiring codes and regulations.



### **Warning**

Risk of electric shock – potential dual supply. Always isolate the power supply/ies to the Heat Battery controller before working on the Heat Battery.



### **Caution**

Each Heat Battery must be protected by its own 16A MCB and have a double pole isolating switch with a contact separation of at least 3mm in both poles, which is located in close proximity to the Heat Battery.



### **Caution**

The Heat Battery must be filled with water and fully vented in both primary and secondary circuits before switching on the Heat Battery electrical supply.



### **Caution**

Use the strain reliefs supplied with the Heat Battery to ensure cables are secured in place.

The following Figure 9 identifies the location of terminals and types of wiring used inside the Flextherm Eco G2 D Heat Battery controller.

The following Figure 9 identifies the location of terminals and types of wiring used inside the Flextherm Eco G2 D Heat Battery controller.

### **Notice**

For installation-specific wiring configurations, please refer to sections 6.4.1-6.4.3. For solar permanent wiring option refer to document FTE-PV.



### Wiring Diagram

C1420-1

WIRING DIAGRAM TYPE 1-R0\_OP - DC P58 EU DSR HP

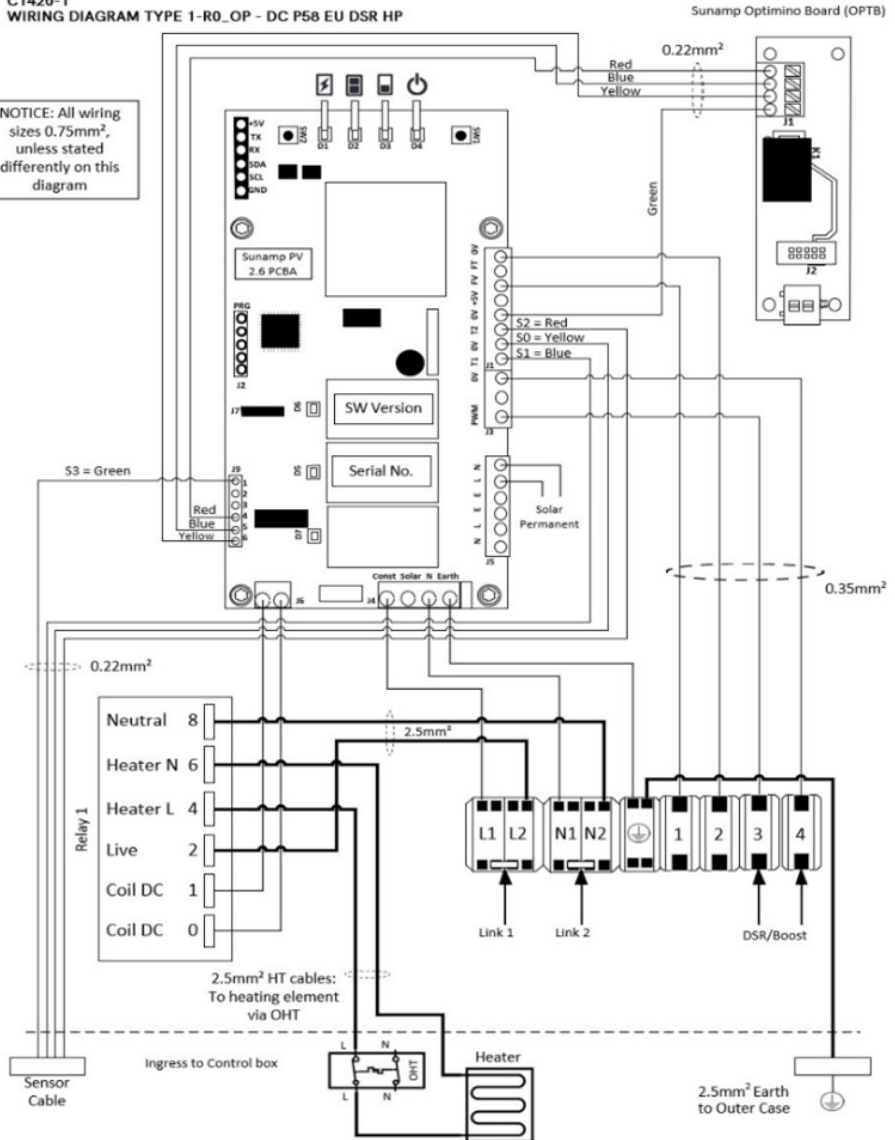


Figure 9 - Flextherm Eco G2 D Heat Battery wiring overview

- (Refer to Figure 10) Move the cable strain relief fittings (1) to the side you wish to make the entries / exits. Cover all other holes in the enclosure with the supplied blanking grommets (2).

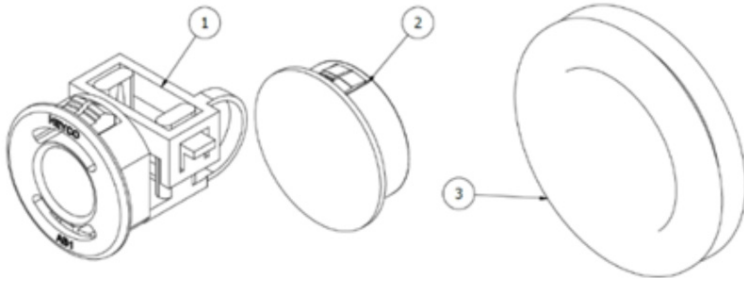


Figure 10 - Flextherm Eco G2 D Heat Battery Grommets & Strain Reliefs

- Identify the Electrical installation type for the Heat Battery
  - Without Solar Power Diversion follow (Section 6.4.1, Figure 11)
  - With Solar Power Diversion follow (Section 6.4.2, Figure 12).
- Feed the mains cable through the strain relief bushing in the Heat Battery housing.
- If applicable, feed any input or output cables through the additional strain relief fittings provided. Refer to Section 6.3.4 for details.
- Line up the cable strain relief fitting ratchet and compress firmly so that the fitting grips the cable.
- Open the internal controller enclosure using a flat head screwdriver to remove the snap in' lid.
- Connect the wires according to the wiring options if not already fitted.
- Remove or make links according to the wiring options.
- Close the internal controller enclosure by replacing the 'snap fit' lid.
- Ensure that the Heat Battery is earthed correctly by checking that the earth clamp on the copper pipe is secure.

#### 6.4.1. Installation without solar power diversion

When using the Flextherm Eco G2 D Heat Battery with a 24/7 mains power supply, the following wiring setup is required. Please refer to Section 6.4.3 for optional control elements, such as timers or boost buttons.

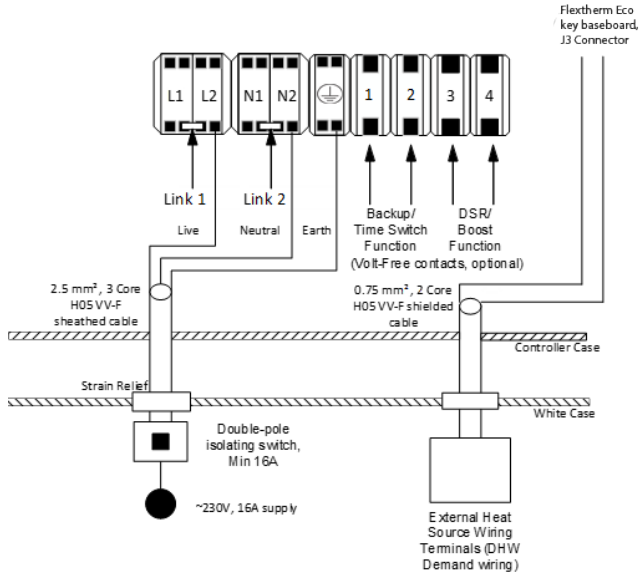


Figure 11 - Flextherm Eco G2 D Heat Battery wiring without solar power diversion

#### 6.4.2. Installations with solar power diversion



##### Warning

Potential dual-supply – Always isolate both supplies to the Heat Battery controller before working on the appliance. Ensure that Links 1 & 2 are removed (see Figure 12) and that the power diversion controller output is modulating AC. Modulating DC power diversion controllers are NOT compatible with the Heat Battery and their use poses a fire risk.

##### Notice

To set the Flextherm Eco G2 D Heat Battery into PV mode, insert a compatible Flextherm Eco G2 D heat pump key into connector J2 on the Flextherm Eco G2 D heat pump key baseboard (see Figure 9). This will change the control strategy of the Flextherm Eco G2 D Heat Battery to call for heat from the heating element connected to a power diversion controller to maximise the self-consumption of PV electricity in the system. This control mode should not be used with a 24/7 grid supply, as it can lead to increased electricity demand.

A list of Flextherm Eco G2 D heat pump keys compatible with the Flextherm Eco G2 D range can be found on <https://flamco.aalberts-hfc.com/nl>.

Follow your chosen power diversion controller's installation manual and document FTE-PV for wiring instructions to the power diversion controller.

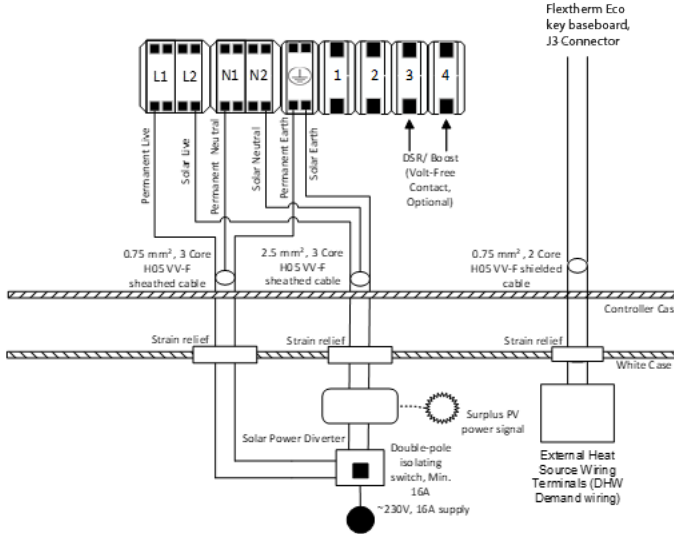


Figure 12 - Flextherm Eco G2 D Heat Battery wiring with solar power diversion

### 6.4.3. Optional Control Inputs & Outputs



#### Caution

Applying excessive downward force onto the PCBA can break the soldering joints. Always remove plug connectors before adding or removing wiring to the screw terminals.

Wiring requirements of the optional inputs & outputs of the Heat Battery can be found in Table 8. The operation details of the inputs & outputs can be found in Section 8.

| Function (I/O)               | Type                    | Location                            | Cable Specification                      |
|------------------------------|-------------------------|-------------------------------------|------------------------------------------|
| DSR/ Boost (Input)           | Volt-Free / Dry Contact | J3 "O" & "PWM" screw terminals      | H05 VV-F, 0.75mm2, 2 core sheathed cable |
| Backup/ Time Switch* (Input) | Volt-Free / Dry Contact | Orange Block Connectors "1" & "2"   | H05 VV-F, 0.75mm2, 2 core sheathed cable |
| Solar Permanent (Output)     | ~230VAC, 3A             | J5 "HEAT" "N" & "L" screw terminals | H05 VV-F, 1.5mm2, 2 core sheathed cable  |

|                                                   |                            |                                         |                                          |
|---------------------------------------------------|----------------------------|-----------------------------------------|------------------------------------------|
| External Heat Source DHW Demand Sensor (Output)** | Volt-Free/ Dry Contact**** | Flextherm Eco G2 Baseboard J3 Connector | H05 VV-F, 0.75mm2, 2 core shielded cable |
|---------------------------------------------------|----------------------------|-----------------------------------------|------------------------------------------|

**Notes:**

\*This function is only available when using the Flextherm Eco G2 D Heat Battery without a solar power diversion controller. Please note that engaging this contact will lead to charging from the internal electric backup heater, which can lead to increased running costs.

\*\* Please refer to the External Heat Source Manual for wiring configuration on External Heat Source controller. If necessary, please use the DHW sensor provided with the external heat source and amend wiring to insert into Flamco Flextherm Eco G2 D Key Baseboard (OPTB) J3 connector. For more information refer to document FTE-HP.

\*\*\* This function is only available for Flextherm Eco G2 D Heat Batteries with a solar power Diversion controller, for more information on this, please refer to document FTE-PV.

\*\*\*\* If using in Hot Water Thermostat mode with HP-Key or PV+HP-Key Flextherm Eco G2 D Key, the contact can accept a ~230VAC, 3A max or 12-24 VDC, 0.5A max switching signal.

Table 8- Flextherm Eco G2 D Controller Inputs & Outputs

6.4.4. Installation of the Flextherm Eco G2 D Key

To install the Flextherm Eco G2 D Key, isolate the Heat Battery from the power supply and refer to the section 6.4. A graphical description of the installation process is provided with the key.

## 7. Commissioning

### 7.1. General

**Caution**

Before commissioning the Heat Battery, first check that you have properly reviewed the previous sections, particularly regarding Heat Battery specifications as well as location, electrical supply and water supply requirements.

**Caution**

The Heat Battery and its associated pipework **MUST** be filled and fully vented in both Primary and Secondary circuits before the Heat Battery power supply is switched on.

### 7.2. Pre-commissioning checklist

- Check all packaging material has been removed.
- Check all components are clean and undamaged.
- Adjust PRV on Cold Mains Supply if the Pressure exceeds 5 Bar (0.5MPa).
- If present, adjust the flow regulator valve to within the maximum recommended flow rate for the Heat Battery size installed.
- If required, adjust the Auto-bypass valve on the primary circuit to the recommended flow rate for the Heat Battery size installed.

### 7.3. Commissioning process

1. Check that the temperature sensor of the Heat Battery has not become dislodged during transport and is fully inserted into its pocket. The white marker should be sitting on top of the blue cable gland (see Figure 7).
2. Apply the relevant Flextherm Eco G2 Key to the Flextherm Eco G2 Key Baseboard, by following the instructions on the manual provided with the Flextherm Eco G2 Key. Ensure that the Key is relating to the External Heat Source installed, if unsure, please contact Flamco.
3. Turn on the water supply and check that there are no leaks.
4. Fully open any hot water taps in the dwelling and allow it to run for a minimum time of 2 minutes. This is for any air to leave the system. This may vary depending on Heat Battery model size.

**Notice**

For larger models such as the Flextherm Eco 9D & 12D G2 the minimum flush time should be increased to 4 minutes.

5. Switch on the power supply to the Heat Battery, External Heat Source and solar power diversion controller (if fitted).

6. Check the front of the Heat Battery to ensure that the “power” LED is lit (see Figure 13 & Table 9).
7. Apply the relevant DHW settings for the External Heat Source (For settings for compatible heat pumps, please check ‘FTE-HP+PV’).
8. Observe on the External Heat Source controller interface if the DHW demand has been activated and the External Heat Source Diverter Valve has moved into DHW position.
9. If present, press the BOOST button on the solar power diversion controller to start charging the Heat Battery. The operation of the External Heat Source in DHW mode, will be paused and the Heat Battery heating element will engage, observe if the “heating element” LED is lit (see Figure 13 & Table 9).
10. Allow the Heat Battery to charge for approximately 30 minutes with the hot water tap closed.
11. If using the heating element for charging, eg when using a PV key, please note that on first charge or when the Heat Battery has been switched off and cooled down the heating element will cycle ON and OFF for up to one hour, depending on the Heat Battery size. This is normal operation. If heating element cycling persists for more than one hour, please refer to (Table 10).
12. If fitted with a Solar Power Diverter, please apply a setting to ensure that the Heat Battery does not stop charging while it is cycling during start up. Please refer to the Solar Power Diverter Manual.
13. After 30 minutes open the hot water tap/s and check for hot water.
14. Adjust the Hot Water Tempering Valve, so that the output temperature is 45°C to 55°C.
15. Check hot water temperature at all hot water outlets in the dwelling with the customer and advise on temperature settings.
16. Ensure that the Heat Battery charges to half charge and that there are no lights flashing (which may otherwise indicate an error, Table 10).
17. If the Heat Battery is fitted with a Solar Power Diverter - Pre-planned Boost times should be set on the Solar Power Diverter, the information on how these are set can be found in the relevant section of the Solar Power Diverter manual. These depend on the end user usage type of the system.
18. Once the installation is finished, please carry out the following steps:
  - Explain to the customer / end user all safety precautions.
  - Fill in and return the Commissioning certificate provided with the Heat Battery. These documents MUST be completed and returned to Flamco after the installation to obtain the extended warranty.
  - Leave all product information and literature with the customer / end user.
  - It's the responsibility of the end user to supply this manual to any other subsequent users.

### **Final Fit After Commissioning:**

Commissioning instructions are provided in this Section 7 of the manual. Follow the instructions below after commissioning.

- Cut the 32mm thick insulation layer to suit the pipe and cable entries. This layer has several perforations for guidance. This can be done with a sharp knife or scissors. Please DO NOT cut inside the Heat Battery and near the Vacuum Insulation panels located at the side of the Heat Battery.
- Re-place the newly cut 32mm thick insulation layer nesting the insulation around the pipe work and cables. Makes sure the main and signal cables sit above this layer.
- Re-place the top 10mm layer of insulation.
- Re-place the lid, aligning the rear pins with the slots in the rear of the unit, slide back, fit the 2 x M5 button head cap screws using 3mm hex head.
- Fix any Energy labels provided in the document pack to product main body.



## 8. Operation

### LED operation

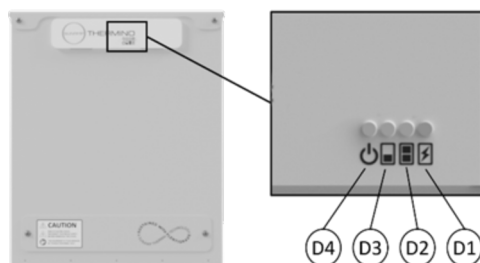


Figure 13 - Flextherm Eco G2 D Heat Battery LED operation

| LED                                                                                | Function                       | Status       | Operation Description                |
|------------------------------------------------------------------------------------|--------------------------------|--------------|--------------------------------------|
|   | Mains Power LED (D4)           | OFF          | Power OFF                            |
|                                                                                    |                                | ON           | Power ON                             |
|   | Charge Level Status 1 (D3)     | OFF          | Heat Battery ON - no charge demand   |
|                                                                                    |                                | Pulsing      | Heat Battery charging from 0 - 50%   |
|                                                                                    |                                | ON           | Heat Battery charge level >50%       |
|   | Charge Level Status 2 (D2)     | OFF          | Heat Battery charge level 0 - 50%    |
|                                                                                    |                                | ON - Pulsing | Heat Battery charging from 50 - 100% |
|                                                                                    |                                | ON - Solid   | Heat Battery charge level 100%       |
|  | Heating Element Operation (D1) | OFF          | Heating element INACTIVE             |
|                                                                                    |                                | Solid        | Heating element ACTIVE               |

Table 9- Flextherm Eco G2 D Heat Battery LED operation

**The following settings detail the operation of the additional wiring functions available in Section 6.4.**

### DSR/Boost operation

This function forces the Heat Battery to charge or “top up” by using the integrated heating element, when there is a DSR demand to allow the Heat Battery to maximise the use of low-cost tariffs. Please note that the input to the Heat Battery controller **MUST** be Volt-Free for this function.

### **Solar Permanent operation**

This function enables the Heat Battery to send a charging demand signal to the AC solar power diversion controller. Please note that the Output for this feature is ~230VAC, 3A Max. Refer to the installation instructions of your AC solar power diversion controller for the use of this signal. This option is **ONLY** available for the Heat Battery options with Solar Power Diversion controller. Please refer to document FTE-PV for more information on this function.

### **Backup/ Time Switch Operation**

This function allows the Heat Battery to be set into electric back-up mode. Please note that the charging in back-up mode will exclusively happen via the Heat Battery integrated heating element. The Heat Battery will not charge from the External Heat Source, when this function is engaged. The input to the Heat Battery controller **MUST** be Volt-Free for this function.

### **External Heat Source DHW demand**

This function allows the External Heat Source to connect to the Flextherm Eco G2 D key baseboard, which controls the charging demand to the External Heat Source. The 2-core shielded cable **MUST** be connected to the External Heat Source wiring controls interface DHW tank sensor connector (Volt-free function) or Hot Water Thermostat with HP-Key VoltFree (23017) or PV+HP-Key VoltFree (23025) (~230V AC, 3A max or 12-24V DC, 0.5A max).

## 9. Maintenance



### Caution

Where undertaking maintenance, repairs or removals, ensure that the system is first dis-connected from the electrical and/or water supply.

- In areas where the mains water hardness can exceed 150 ppm Total Hardness and a scale-reducing device has been fitted, the service and maintenance requirements of this device (especially re-fill requirements) need to be adhered to.
- The air pressure in the expansion vessel **MUST** be checked and topped up in line with the expansion vessel manufacturer's instructions for maintenance or annually, whichever is sooner.
- The peripherals and accessories that are part of the Heat Battery install **MUST** be maintained according to their manufacturer's instructions.
- Except as identified above, the Heat Battery does **NOT** require any regular maintenance.
- If the supply cord is damaged, it must be replaced by the manufacturer, its service agent or similarly qualified persons in order to avoid a hazard. Please refer to the Electrical wiring sections.

## 10. Troubleshooting

| Fault                                                          | Possible Cause(s)                                                                                                                               | Possible Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The Heat Battery does not deliver hot water after installation | Issue with External Heat Source                                                                                                                 | Refer to External Heat Source manual. If using a Flextherm Eco G2 D, the DSR/ Boost or back-up switch can be used to supply hot water in this instance. If using a Flextherm Eco G2 D with Solar Diversion Control, the boost button on the solar power diverter will activate the heating element inside the Heat Battery (given there are no issues with the power supply to the Heat Battery or Solar Power Diverter). Please note this will result in higher electricity costs if this is activated in a continuous mode.                                                                 |
|                                                                | Heat Battery not powered or not powered correctly                                                                                               | Check wiring and power supply to Heat Battery and rectify where necessary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                | Solar Power Diverter not powered or not powered correctly (please note this ONLY applies if Heat Battery installed with a Solar Power Diverter) | Check wiring and power supply to Solar Power Diverter and rectify if necessary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                | Non-self-resetting thermal cut-out has tripped                                                                                                  | <p>Make sure the Heat Battery is full of water and purge if necessary:</p> <ol style="list-style-type: none"> <li>1. Disconnect the power supply</li> <li>2. Open the electrical cover at the base of the Heat Battery (Figure 1 -RHS image)</li> <li>3. Reset the non-self-resetting thermal cut-out on Left Hand side of the panel.</li> <li>4. Check if the temperature sensor is fully inserted in the Heat Battery (Section 7.3)</li> <li>5. Reassemble the Heat Battery, and</li> <li>6. Reconnect the supply.</li> </ol> <p><b>If the problem persists, please contact Flamco.</b></p> |
| LED D4 (power symbol) is rapidly flashing                      | Temperature sensor string is faulty                                                                                                             | <p>Check that the sensor cable is properly connected to the PCBA and that the connector block or screw terminal is making good contact.</p> <p><b>If the problem persists, please contact Flamco.</b></p>                                                                                                                                                                                                                                                                                                                                                                                     |
| The flowrate from the Heat Battery is lower than expected      | Air might still be trapped in the system                                                                                                        | Check that the mains supply valve is fully opened. Make sure the Heat Battery is fully de-aired and purge if necessary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                | Incoming water pressure is too low                                                                                                              | Measure the incoming pressure from the mains supply and <b>contact Flamco.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Table 10 - Troubleshooting of Flextherm Eco G2 D Heat Battery

# 11. Decommissioning & disposal

## 11.1. Decommissioning

To successfully decommission the Heat Battery, please carry out the following steps:

1. Switch off all Electrical Power Supplies to the Heat Battery.
2. If Heat Battery is charged and no leaks are present, cool down the Heat Battery by flowing cold water through, until the temperature at the Outlet side is equal to the temperature at the Inlet.
3. Isolate the Cold Mains Supply to the Heat Battery.
4. Open the hot water taps to drain the secondary circuit system and release the pressure in the pipe work.
5. Drain the primary circuit that is connected to the External Heat Source.
6. Remove all electrical cables and connections from the Heat Battery controller.
7. Remove all pipework from the Heat Battery connections, using appropriate tools and methods. Cap pipework if it cannot be removed from the Heat Battery connections.
8. Ensure that the Heat Battery is left to cool for at least 60 minutes after completion of step 2 before moving it.

## 11.2. Disposal



This symbol on the Heat Battery and accompanying documents means the Heat Battery should not be mixed with general household waste at the end of its life.

For proper treatment, recovery and recycling, please take the Heat Battery to designated collection points where it will be accepted free of charge at the end of its life.

Disposing of this Heat Battery correctly will help save valuable resources and prevent any potential negative effects on human health and the environment, which could otherwise arise from inappropriate waste handling.

Please contact your local authority for further details of your nearest designated collection point. Penalties may be applicable for incorrect disposal of this waste, in accordance with your national legislation.

## 12. Complementary products



**Airfix**  
Potable Water  
Expansion Vessels



**Flexofit Super**  
Water Shock Arrestor



**Flamcomix**  
Thermostatic Mixing Valve



**Prescor IC**  
Safety Group



**Prescor B**  
Boiler Valve



**Prescor PRV**  
Pressure Reducing Valve



Isolation Valve



Isolation Valve

Find out more on: <https://flamco.aalberts-hfc.com>



## Contact us !

We supply products for the installation industry in more than 70 countries. This is done from Flamco sales offices and via distributors who know the local market, and can give you the right advice at all times.

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